

Education

- 2015-2020 **Dual Degree (B.Tech + M.Tech) in Computer Science and Engineering**
Indian Institute of Technology Madras, Chennai, India CGPA: 8.78
- 2015 **XII - Karnataka Board, KLE Society's Independent PU College, Bangalore** 97.30 %
- 2013 **X - ICSE, B P Indian Public School, Bangalore** 96.33%

Professional Experience

- Software Engineer, Level IV, Google LLC, New York**
- Dec 2022 - Present **Software Engineer, Level IV, Google India Pvt Ltd, Bangalore**
- Aug 2020 - Nov 2022 { Developing intelligent features for the Google Workspace Editors (Docs, Slides, Keep, etc) using my expertise on the products' client-side software, supporting tools and libraries, and natural language processing infrastructure.
 { Using cutting-edge frontend tools like Web Assembly and Emscripten, and Google-internal technologies like j2Cl, client-side cross-platform frameworks and build systems, to develop user-facing features such as spellcheck in encrypted documents for five languages and writing style suggestions for English text.
 { Formulating technical designs for independent end-to-end problems, driving cross-team collaboration, upholding software reliability practices, technical-debt resolution and documentation, and proactively identifying areas of future work.
 { Guiding junior engineers on programming and software design tasks to enable timely delivery of products to customers.
- May 2019 - July 2019 **Software Engineering Intern, Google India Pvt Ltd, Bangalore**
- Worked on the Editors client-side software infrastructure to develop a user interface with control options to undo or provide feedback on the correction and a logging framework, for the Google Docs text auto-correction feature.
- May 2018 - July 2018 **Research Intern, Big Data Experience Labs, Adobe Research, Bangalore**
- Developed a mobile application for Text to Scene Conversion in Augmented Reality, based on novel research techniques for prediction of three-dimensional object sizes and positions from textual features.

Research Experience

- Sep 2019 - May 2020 **Paraphrase Generation with a Bilingual Model and Continuous Embeddings** *Master's Thesis, Language Technologies Institute, Carnegie Mellon University*
- Machinated a novel technique for paraphrase generation using the [von Mises-Fisher \(vMF\) Loss](#) on a transformer network, and showed that it produces superior paraphrases as compared to the log-likelihood model by employing bilingual data to induce zero-shot paraphrasing, guided by [Prof. Yulia Tsvetkov](#).
- May 2017 - July 2017 **Cognitive Approach to Natural Language Processing**
- Research Intern, Department of Computer Science and Automation, Indian Institute of Science (IISc), Bangalore*
- Developed a cognitive text parser that combines syntactic and semantic approaches, to process textual data into cognitive structural representations, to be used as a feature extractor for downstream NLP tasks, and demonstrated the correlation of the extracted cognitive features with semantic and syntactic text features, guided by [Prof. Veni Madhavan](#).

Publications and Patents

- [Publication and Poster] **Improving the Diversity of Unsupervised Paraphrasing with Embedding Outputs (Paper, Poster)**
Monisha Jegadeesan, Sachin Kumar, John Wieting, Yulia Tsvetkov
 In [Workshop on Multilingual Representation Learning](#),
 The 2021 Conference on Empirical Methods in Natural Language Processing ([EMNLP 2021](#))
- [Publication and Poster] **Adversarial Demotion of Gender Bias in Natural Language Generation (Paper, Poster)**
Monisha Jegadeesan
 In [ACM CODS-COMAD 2020](#) - Young Researchers' Symposium
- [Poster] **ARComposer: Authoring Augmented Reality Experiences through Text (Poster)**
Sumit Kumar, Paridhi Maheshwari, Monisha Jegadeesan, Amrit Singhal, Kush Kumar Singh, Kundan Krishna
 In ACM User Interface Software and Technology Symposium 2019 ([ACM UIST 2019](#))
- [Filed Patent] **Visualizing Natural Language through 3D Scenes in Augmented Reality**
Sumit Kumar, Paridhi Maheshwari, Monisha Jegadeesan, Amrit Singhal, Kush Kumar Singh, Kundan Krishna
 Filed at the US PTO (Application Number: 16/247,235)

Projects

Real-Time AI-Powered Customer Support Chatbot (2024)

Designed, trained, and optimized an AI model to build a real-time chatbot for customer support that integrates company knowledge bases using retrieval-augmented generation (RAG) techniques. Built scalable pipelines for model deployment and incorporated vector databases for efficient information retrieval, collaborating with engineering teams to integrate the AI solution into the customer support platform. The system significantly improved response times by providing instant replies and reduced human agent workload by 35%. Proactively monitored and fine-tuned the model performance in production to ensure reliability and scalability.

Tech: LangChain, OpenAI API, ChromaDB, Python

Predictive Maintenance Model for Industrial Equipment (2024)

Developed and optimized a predictive maintenance machine learning model leveraging sensor data to forecast equipment failures and reduce unplanned downtime. Constructed data preprocessing and feature engineering pipelines for time-series IoT sensor data including temperature, vibration, and pressure metrics. Trained Random Forest classifiers and implemented rigorous model validation using F1-score and precision metrics to ensure reliable production-ready deployment. Collaborated with data engineers to build scalable model deployment pipelines and monitored model performance to continuously improve operational efficiency and cost savings.

Tech: Python, Random Forest, Sensor Data Processing

Customer Sentiment Classification Using NLP (2024)

Engineered an end-to-end machine learning pipeline to classify customer sentiment from e-commerce product reviews by designing feature extraction and preprocessing workflows including text cleaning, tokenization, and TF-IDF vectorization. Built and optimized a Logistic Regression model to deliver high-accuracy sentiment predictions, enabling product teams to identify key complaint themes for targeted improvements. Worked in close collaboration with data and product teams to integrate the AI model into analytics platforms, ensuring maintainable, well-documented code and robust validation through model evaluation techniques.

Tech: Python, NLTK, TF-IDF, Logistic Regression

Sales Forecasting Model for Retail Inventory Optimization (2024)

Designed and deployed time series forecasting models to predict monthly retail sales, supporting inventory planning and revenue optimization goals. Conducted comprehensive feature engineering including time-series decomposition and seasonality analysis to enhance model accuracy. Implemented ARIMA and Facebook Prophet algorithms, building scalable training and validation pipelines. Collaborated with cross-functional teams to integrate forecasting outputs into inventory management systems, successfully reducing forecasting errors by 25% and optimizing stock control. Maintained rigorous model monitoring and continuous performance improvements post-deployment.

Tech: Python, Prophet, ARIMA, Pandas

About :

I specialize in designing, training, and optimizing scalable and reliable AI and machine learning models to solve real-world problems. Proficient in Python and ML frameworks such as TensorFlow, PyTorch, and scikit-learn, with deep expertise in data preprocessing, feature engineering, model evaluation, and performance monitoring. Experienced in building end-to-end pipelines for model training, validation, deployment, and integration into applications through collaboration with data engineering and product teams. Skilled in leveraging cloud platforms and MLOps practices to ensure smooth production deployment and continuous improvement of AI systems. Committed to writing clean, maintainable, and well-documented code and continuously researching and implementing new AI techniques to drive measurable business impact.

